PAUL VARGHESE SHIBU

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SUMMARY

Mechanical Engineering undergraduate at NIT Calicut with hands-on experience in CFD, fluid-flow systems, propulsion, manufacturing, and experimental testing through internships at ISRO/LPSC, Inbound Aerospace, and Wilo Mather and Platt. Experienced in ANSYS, CAD, CNC manufacturing, and machine learning, with additional experience leading multidisciplinary technical teams.

EDUCATION

National Institute Of Technology Calicut — Aug 2023 - May 2027
B.Tech in Mechanical Engineering (Minor in Robotics) — *CGPA: 8.23*

St Antony's Public School — July 2020 - Aug 2022
Higher Secondary Education (Kanjirappally) — *Percentage: 92.8%*

Nirmala Public School — Until 2020
High School — *Percentage: 96.4%*

WORK EXPERIENCE

ISRO / Liquid Propulsion Systems Centre (LPSC) — Dec 2025 - Jan 2026
Intern - Earth Storable Engines and Stages (ESES)

- Project: Hot gas Roll Control System(HRCS) of semi-cryogenic stage. (Internship Report)
- Performed compressible CFD analysis of hot gas tapped from pre-burner exhaust through an orifice-nozzle setup using ANSYS.
- Designed and evaluated orifice geometries for controlled hot-gas flow based on pressure drop and mass flow requirements.

Inbound Aerospace | IIT Madras Research Park — May 2026 - July 2026
Spacecraft Propulsion System Design Intern

- Design and development of an N_2O-based green monopropellant reaction control thruster.
- Performed CFD optimization of propellant feed lines to improve flow characteristics and system efficiency.
- Designed and fabricated a ground test rig setup for evaluating pulsed-mode thruster operation. (Test Video)
- Conducted cold-flow ground testing of the pulsed thruster and characterized thrust response using load-cell measurements.

NIT Calicut | ALTIUS SURYA TECH Pvt. Ltd.(Under Dr. Sai Saketha) — June 2025 – August 2025
Machine Learning Intern

- Developed a solar PV energy forecasting pipeline using three years of real world power generation data, applying SARIMAX time series modeling to capture annual seasonality and weather driven variations.
- Integrated PVGIS API for global solar irradiance data and used 'pvlib' to simulate realistic energy output. (Certificate)

Wilo Mather and Platt Pumps Ltd. — May 2025 - June 2025
Mechanical Intern - Machine Shop

- Evaluated multi-axis CNC operations (Widma, Trevisan, You Ji) and verified G/M-code toolpaths for pump impellers.
- Analyzed and simulated CNC code using CAMotics for validation before it goes to production.
- Evaluated production requirements to determine optimal machine routing, trade-offs between setup time and cycle efficiency across CNC lathes and high-end multi-axis machines.

Progressive Software Solutions and Training — May 2024 - Aug 2024
Web Developer Intern

- Built dynamic frontend applications using React.js and backend APIs with Node.js and Express.

PROJECTS

- **Autonomous Drone for Payload Delivery** *(Aerial Robotics/UAV)*
  - **Tech Stack:** Pixhawk Cube Orange+, Raspberry Pi, Pi Cam, OpenCV, Mission Planner
  - Designed and tuned an autonomous quadcopter running ArduPilot on a Pixhawk Cube Orange+ with Raspberry Pi as companion computer (Design Report).
  - Led the mechanical design and 3D printing of custom components and payload dropping mechanism for a quadcopter.
  - Programmed precision release algorithms and target detection using OpenCV and onboard telemetry feeds (Flight Video).
  - Secured **AIR 2** in Payload Mechanism Design and **AIR 6 Overall** at the SAE Autonomous Drone Development Challenge.

- **Vision-Language-Action (VLA) Controlled Robotic Manipulator** *(Final Year Project)*
  - **Tech Stack:** LeRobot (Hugging Face), OpenVLA, pi0, SO-100 Arm (Video Demonstration)
  - Designed and fabricated a low-cost, 5-DOF robotic arm for imitation learning and VLA model deployment.
  - Deployed an end-to-end pipeline via the **LeRobot** ecosystem, mapping natural language and vision to motor actions.
  - Evaluated VLA models, including **pi0** and **OpenVLA**, for spatial reasoning and precise pick-and-place execution.

- **Physics-Informed Neural Networks (PINNs) for CFD** *(Academic Project)*
  - Built PINNs using PyTorch and DeepXDE to solve forward and inverse fluid dynamics problems.
  - Solved the Eikonal equation to compute geometry-agnostic wall distance fields for complex airfoils.
  - Solved 2D Navier-Stokes equations to analyze flow fields and recirculation past an ellipse.
  - Designed an inverse PINN to recover unknown Reynolds numbers from sparse wake-clustered data.

- **Reinforcement Learning (RL) for CFD and Flow Optimization** *(Academic Project - Ongoing)*
  - Developing a mathematical framework integrating Reinforcement Learning (RL) algorithms with CFD simulations to optimize fluid flow characteristics and **active flow control**.
  - Coupled RL agents with **OpenFOAM** to autonomously iterate and evaluate complex flow fields, accelerating the traditional design and simulation cycle.
  - Extracted and analyzed dynamic simulation data to train the model's control policy, validating its ability to achieve targeted aerodynamic performance and pressure distributions.

- **Additive Manufacturing of Autonomous Drone Components** *(Academic Project)*
  - Modeled and 3D-printed custom functional components, including a multi-tiered electronics bay, to convert a Mark 4 7-inch FPV drone frame into an autonomous flight platform (Project Report).
  - Designed aerodynamic propeller ducts to provide crash protection, reduce tip vortices, and actively increase lift.
  - Fabricated components with PLA. Utilized Autodesk Meshmixer for geometry inspection and Ultimaker Cura to optimize support-free toolpaths.

- **Gearless 90° Power Transmission Mechanism** *(Production Engineering Lab)*
  - Designed and fabricated a gearless 90° power transmission mechanism using perpendicular shaft arrangements.
  - Carried out process planning, machining, drilling, welding, and assembly of the base frame and shaft components.
  - Gained experience in manufacturing processes, mechanical assembly, and functional validation of a fabricated transmission system.

- **Fixed-Wing UAV Development & Missions**
  - Secured **3rd Place** at IIST Trivandrum in the Aerobatic Plane Competition, designing for high maneuverability and structural integrity.
  - Developed custom fixed-wing aircraft for payload delivery missions, achieving **5th Place** at National Aeromodeling NITK and **6th Place** at IIT Bombay Aeromodeling Challenge. Flight Video
  - Ranked **8th Nationally for CFD** and 15th Overall at the SAE India Drone Development Challenge (DDC) 2025.
  - Managed full-cycle development from CAD design and fabrication to flight testing.

- **5-inch FPV Quadcopter (6S Architecture)**
  - **Tech Stack:** DreamflyRC F722 FC, GEPRC BLS 50A 4-in-1 ESC, XING 2207 1750KV Motors, SpeedyBee TX800
  - Developed a 5-inch drone on a Mark 4 V2 Carbon Fiber frame, optimizing thrust-to-weight ratio for 6S power systems.
  - Configured flight controller firmware and PID tuning via Betaflight to ensure stability and precision during high-velocity maneuvers.(Drone Image)

- **Wildlife Detection & Alerting System**
  - **Tech Stack:** OpenCV, YOLOv8, Raspberry Pi, Pi Camera
  - Built a system for real-time wildlife detection using YOLOv8.

- Integrated OpenCV for image processing and Raspberry Pi and Pi cam for live video capture and processing.

- **Gold Price Prediction using Ensemble method**
  - Engineered a multi-stage stacking pipeline (LGBM, CatBoost, XGBoost) to classify trading actions (Buy/Hold/Sell).

- **Smart Bicycle Sharing System for Intra-Campus Mobility, NIT Calicut**
  - As part of an elective course on multimodal transportation systems, I co-developed a comprehensive business plan for a smart bicycle sharing system tailored to NIT Calicut's campus.

- **Ecommerce Web App**
  - A full-stack shopping website with a Node.js/Express backend that handles user login, shopping carts, and checkout, built to work with a React front end that's still being developed.

- **Wind Tunnel**
  - Developed a wind tunnel for a college exhibition, showcasing aerodynamic principles and their applications.

MANAGERIAL EXPERIENCE

Club Aerounwired Sept 2024 - Present

Secretary

- Led multidisciplinary teams to top finishes across major national competitions, notably securing **All India Rank 2** for best payload dropping mechanism and **6th overall** at SAE Autonomous Drone Development Challenge (Event).
- Managed custom fixed-wing aircraft development for payload missions, achieving **5th Place** at National Aeromodeling NITK, **6th Place** at IIT Bombay Aeromodeling Challenge, and ranking **8th Nationally** in CFD (**15th Overall**) at the SAE India DDC 2025(Flight Video).
- Organized large-scale, on-campus technical workshops, including Model Rocketry for 300+ attendees and the ICARUS fixed-wing drone workshop for 100+ attendees.
- Led the design and development of the Club Aerounwired website, coordinating the development team from UI design to deployment. (Website)

IEEE SB NITC Feb 2025 - Present

Webmaster (Former Technical Coordinator)

- Conducted 115+ on-campus events, coordinating between 11 societies.
- Managed technical operations for the IEEE YESS 2025 summit which hosted 50K+ participants.
- Designed and deployed the official website for the National Power Electronics Conference (NPEC 2025).
- Led a dedicated web development team to build the new official website for the student branch(Under Development). (Website)

Club Mathematica NITC Nov 2024 – Present

Treasurer

- Conducted INFINITUM, a school-level quiz competition with 28,000 participants across South India.
- Led Felizora and Navaragam events at NITC, handling ideation to execution for 500+ and 600+ attendees respectively.
- Contributed to branding and outreach as part of the club's design and marketing team.

Tathva '24

Tech & Design Team

- Assisted in developing the Tathva '24 website and created event posters and brochures.

SKILLS

Engineering: CFD, CAD, Additive Manufacturing, Aerospace Propulsion

Engineering Software: ANSYS, Fluent, Fusion 360, CAMotics, SolidWorks

Programming: Python, Gazebo, Git, JavaScript

Robotics/UAV: ArduPilot, Mission Planner, DroneKit, PyMavlink, Raspberry Pi

Other: Technical Leadership, Team Management, Event Organization